

Atomic Clock Frequency Ratios with Fractional Uncertainty $\leq 3.2 \times 10^{-18}$

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We report high-precision frequency ratio measurements between optical atomic clocks based on $^{27}\text{Al}^+$, ^{171}Yb , and ^{87}Sr . With total fractional uncertainties at or below 3.2×10^{-18} , these measurements meet an important milestone criterion for redefinition of the second in the International System of Units. Discrepancies in ^{87}Sr ratios at approximately 1×10^{-16} and the Al^+/Yb ratio at 1.6×10^{-17} in fractional units compared to our previous measurements underscore the importance of repeated, high-precision comparisons by different laboratories. A key upgrade from our previous work is the use of a common ultrastable reference delivered to all clocks via a 3.6-km phase-stabilized fiber link between two institutions, enabling better accuracy and stability in frequency transfer. Derived from a cryogenic single-crystal silicon cavity, this reference improves comparison stability by a factor of 2–3 over previous systems, with an optical lattice clock ratio achieving a fractional instability of 1.3×10^{-16} at 1 s. Identifying individual clock stabilities with a three-corner-hat measurement, we demonstrate the most stable optical lattice and trapped-ion clocks used in a multispecies comparison. By enabling faster comparisons, this stability will improve sensitivity to nonwhite noise processes and other underlying limits of state-of-the-art optical frequency standards.

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Introduction—Frequency references based on spectroscopy of atomic transitions are the core of high-precision timekeeping, enabling global navigation satellite systems, network synchronization, and other technologies central to modern life. The International System of Units (SI) defines

the second based on a 9.2 GHz hyperfine transition in Cs, with the best realizations of this definition having fractional systematic uncertainties at the 10^{-16} level [1–3]. Using frequencies at least 10^4 times higher, optical atomic clocks achieve instability and systematic uncertainty 2 orders of magnitude lower than the best microwave standards [4]. This motivates an effort to redefine the SI second in terms of one or more optical atomic clock frequencies [5]. High-precision frequency ratio measurements are a critical element of this effort, necessary to rigorously demonstrate optical clock performance and repeatability as well as optical frequency dissemination and timescale generation [6–10].

Beyond timekeeping, optical clock comparisons are among the most precise measurements [11,12]. This precision makes them well suited as probes for weak couplings introduced by physics beyond the standard model (BSM) [13], with atomic clock ratios setting limits on potential

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BSM phenomena including various dark matter models [11,14,15], violations of local position invariance [16], and time variation of fundamental constants [17,18]. Frequency comparisons can also enable relativistic geodesy, using frequency shifts between clocks to determine gravitational potential differences [19,20].

We present frequency ratio measurements between three different optical frequency standards, all located in Boulder, Colorado: the $^{27}\text{Al}^+$ clock [21] and ^{171}Yb optical lattice clock (OLC) [22] located at the National Institute of Standards and Technology (NIST), and the ^{87}Sr OLC [23] located at JILA. These three optical clock species share similar electronic structures including narrow (< 10 mHz), environmentally insensitive $^1S_0 \rightarrow ^3P_0$ clock transitions. A frequency comparison between these three species was previously completed in 2021 [11] (BACON21). Since this measurement, the $^{27}\text{Al}^+$ and ^{87}Sr clocks and the frequency distribution network have been rebuilt and undergone new systematic evaluations [21,23]. The ^{171}Yb clock operates largely as previously evaluated [22], with updated evaluations of certain systematic effects presented in Supplemental Material [24].

The optical local oscillators of all three clocks are phase locked to the JILA cryogenic silicon laser reference cavity (Si cavity) [48,49] via a 3.6-km optical fiber network. This reduces comparison instability, enabling statistical uncertainty $< 5 \times 10^{-18}$ for each frequency ratio in 2 h of averaging time. We report the three frequency ratios with total fractional uncertainties $\leq 3.2 \times 10^{-18}$. In addition to being the most precise measurements of any physical quantity (other than symmetry tests, e.g., [50]), the ratios reported in this Letter meet the 5×10^{-18} threshold identified in the road map for redefinition of the SI second [5,12].

Optical network—Figure 1 shows a simplified schematic of the phase-coherent optical fiber network allowing frequency transfer between NIST and JILA [51]. $1.5 \mu\text{m}$ light stabilized to the Si cavity is transmitted to NIST via one fiber branch of the network. An optically encoded maser signal [52] is transferred from NIST to JILA through the same fiber with a wavelength division multiplexer, providing a common microwave reference between laboratories.

Each clock laser is prestabilized to local ultralow expansion (ULE) glass cavities and then stabilized at low bandwidth (< 10 kHz) to frequency combs locked to the Si-stabilized $1.5 \mu\text{m}$ light [49,53–55]. This process transfers the low instability of the Si cavity to the three clock lasers. This reduces the Dick effect noise on the OLC comparison instability, and yields longer atom-light coherence for longer spectroscopy times and reduced quantum projection noise (QPN) in the $^{27}\text{Al}^+$ clock [21].

Measurement results—Between January 16 and March 21, 2025, all three clocks were compared nine times, and ^{171}Yb and ^{87}Sr OLCs were compared on four additional days. The results are presented in Fig. 2. In Fig. 2(a), we show the overlapping Allan deviation of the fractional

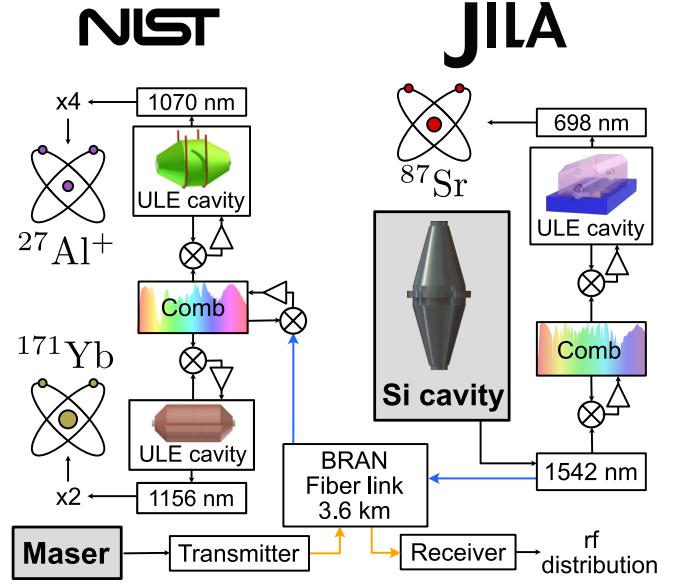


FIG. 1. Simplified schematic of the Boulder optical clock network. Two independent oscillators—a cryogenic silicon (Si) optical cavity and a hydrogen maser for the radio frequency—are distributed across multiple laboratories via the 3.6-km-long Boulder Research and Administration Network (BRAN) optical fiber link. The $^{27}\text{Al}^+$ and ^{171}Yb clocks are located at NIST, and the ^{87}Sr clock is located at JILA. Local ULE cavity-referenced lasers in all laboratories are steered by a common 1542 nm laser locked to the Si cavity at JILA. Frequency combs at JILA and NIST transfer the Si cavity’s stability to the clock wavelengths. A stable microwave signal from a maser at NIST is distributed optically to JILA.

ratios for a single day. A white frequency noise fit [$\sigma(\tau) = \sigma_0/\sqrt{\tau/s}$ with the averaging time τ] to times after 100 s (the servo attack time of the $^{27}\text{Al}^+$ clock) is plotted as a solid line for each ratio. The statistical uncertainty is obtained by extrapolating to the total measurement time for each day. Together with the mean value, this provides one data point for each day’s bin, as plotted in Fig. 2(b).

The average of $\sigma(\tau)$ over the campaign provides an estimate of the instability. The average fractional instability for the frequency ratio between ^{171}Yb and ^{87}Sr (Yb/Sr) is $1.3 \times 10^{-16}/\sqrt{\tau/s}$, which is limited by technical noise and represents more than a factor of 2 improvement over BACON21. The frequency ratios involving $^{27}\text{Al}^+$ with ^{87}Sr (Al^+/Sr), and with ^{171}Yb (Al^+/Yb), are dominated by single-ion QPN, with average fractional instability of $3.9 \times 10^{-16}/\sqrt{\tau/s}$ for both, a factor of 3 improvement, corresponding to a nearly tenfold reduction in measurement time (for a given statistical uncertainty) compared to BACON21.

The high stability of OLCs reveals excess scatter in Yb/Sr with the reduced χ -squared statistic $\chi^2_{\text{red}} = 6.4$, suggesting uncharacterized shifts. While this statistic is comparable to that of BACON21, the between-day

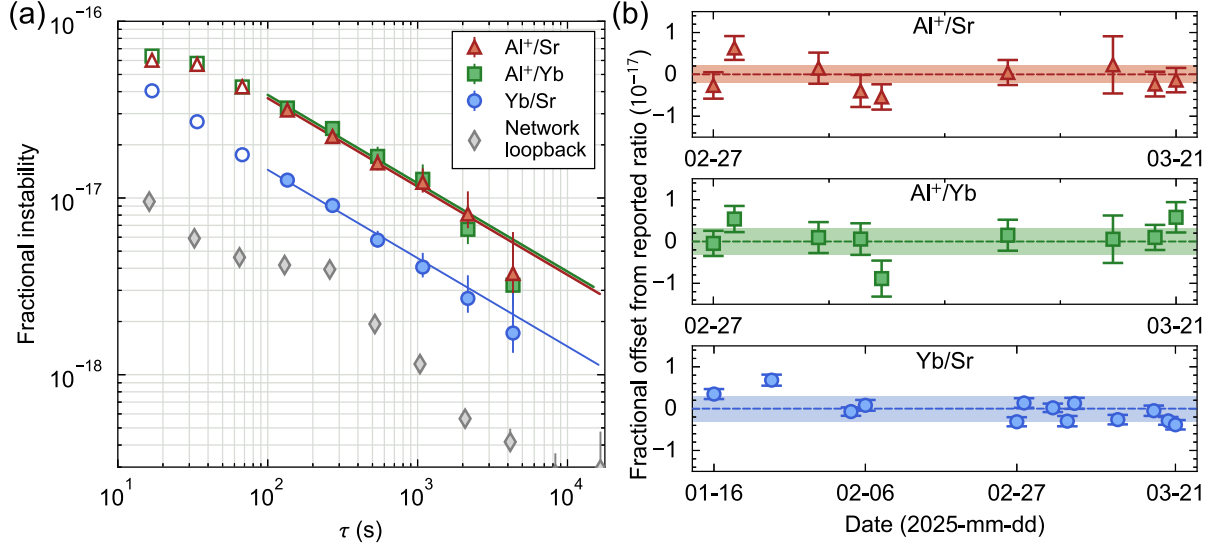


FIG. 2. Frequency ratio measurement results. (a) Fractional instability of the ratio measurements. The points are the overlapping Allan deviation of each fractional ratio, calculated for different averaging times τ and using concatenated data from a single measurement day. The solid lines are a white-frequency-noise model fit to points with τ greater than the servo attack time of ~ 100 s (color filled points). We extrapolate this fit to the total measurement time (the end of the fitted line) to obtain the statistical uncertainty. The gray diamonds are an upper bound on the instability of the fiber link, estimated by a loopback measurement. (b) Daily ratio measurements are shown as fractional deviations from the final reported values [Eq. (1)]. The error bars represent statistical uncertainty obtained by the procedure described in (a). The colored shaded regions show the final fractional uncertainties including systematic uncertainties and are 2.2×10^{-18} , 3.2×10^{-18} , and 3.1×10^{-18} for Al^+/Sr , Al^+/Yb , and Yb/Sr ratios, respectively. All uncertainties represent the 68% confidence interval.

variability is substantially reduced from $10.8(4.2) \times 10^{-18}$ in 2021 to $3.3(9) \times 10^{-18}$ in the present case [24]. The Al^+/Yb and Al^+/Sr ratios show χ^2_{red} values of 1.3 and 1.2, respectively.

We use a comprehensive Bayesian model [11] to aggregate data binned by measurement day, together with systematic uncertainties. The frequency ratios and their standard uncertainties are

$$\begin{aligned} \text{Al}^+/\text{Sr} &= 2.611\,701\,431\,781\,462\,7101(58), \\ \text{Al}^+/\text{Yb} &= 2.162\,887\,127\,516\,663\,6674(70), \\ \text{Yb}/\text{Sr} &= 1.207\,507\,039\,343\,337\,7230(37). \end{aligned} \quad (1)$$

These represent fractional uncertainties of 2.2×10^{-18} , 3.2×10^{-18} , and 3.1×10^{-18} for Al^+/Sr , Al^+/Yb , and Yb/Sr , respectively. Supplemental Material presents a detailed discussion of individual clock systematic uncertainties and the data analysis procedure, and also includes a simpler analysis with similar results [24].

Discussion—All three ratios differ from the values measured in BACON21 and the values recommended by the International Committee on Weights and Measures (CIPM) [56], as shown in Fig. 3(a). The Al^+/Yb ratio differs by 1.6×10^{-17} or 2.4σ from BACON21 in units of combined uncertainty σ . The disagreement in the ^{87}Sr ratios is 14σ , indicating a revision of the ^{87}Sr frequency by approximately 1×10^{-16} . With recently improved

evaluation of atomic coefficients, we could retroactively adjust the BACON21 results. The Sr and Yb frequency would increase by 7.5×10^{-18} and 1.5×10^{-18} , respectively [23,24], and this would increase the deviation from (14.5, 2.4, 13.9) to (15.4, 2.6, 14.7) in units of σ for (Al^+/Sr , Al^+/Yb , Yb/Sr).

The ^{87}Sr clock was rebuilt with several improvements over the previous generation [59]. The thermometers for blackbody radiation (BBR) characterization are mounted on an in-vacuum translational stage, allowing calibration of the BBR shift before and after daily ratio data collection. The atomic density shift is corrected *in situ* using local frequency imaging, and further suppressed by operating at a shallow optical lattice near the magic depth, where the collisional shift is minimized [60]. The lattice light shift and dc Stark shift were reevaluated before and after the campaign to bound potential month-scale changes, and no significant drift was observed.

The $^{27}\text{Al}^+$ clock was similarly rebuilt and improved over the previous generation [61], with reduced excess micro-motion and improved vacuum pressure. The stability of the Si cavity, transferred to the $^{27}\text{Al}^+$ clock laser, allows an extension of the probe time from 150 ms to 1 s, requiring continuous Doppler cooling to control heating of the trapped ion's secular motion. For full details of each clock's operation, see Supplemental Material [24].

We noticed the discrepancy with BACON21 values in 2024 test runs and developed protocols to verify the accuracy

of both the optical network and a number of systematic corrections for each clock. The JILA laser stability transfer system supports 10 different experiments, adding complexity in the network. For this campaign, we therefore implement a new loopback test scheme [see Fig. 3(b)] to evaluate the end-to-end accuracy and stability of the frequency network, including every element from Si cavity to ^{87}Sr atoms. The Si cavity-stabilized $1.5\text{ }\mu\text{m}$ light is sent to NIST, where it references a frequency comb. The comb then stabilizes a 1397 nm laser, which is sent back to JILA in a parallel fiber. At JILA, this light is frequency doubled to 698 nm and compared against the ^{87}Sr clock laser, which is stabilized to a JILA comb referenced to the Si cavity, picked off just before the atoms. The resulting beat note indicates fractional instability $4.2 \times 10^{-17}/\sqrt{\tau/s}$, plotted as diamond points in Fig. 2(a), and a fractional frequency offset of $0.9(1.6) \times 10^{-19}$. Since the light travels through two 3.6-km fibers, we take this measurement as an upper bound on the network instability and inaccuracy.

certain systematic effects while the other two operated under nominal conditions. Systematic effects tested include the environment temperature, magnetic bias field magnitude, lattice depth, and spectroscopy time of the ^{87}Sr clock; the Doppler cooled motional energy of the $^{27}\text{Al}^+$ clock; and the magnetic bias field, measurement cycle time, and environment temperature of the ^{171}Yb clock. Supplemental Material [24] includes discussion of each modulated systematic effect, as well as any significant differences in operational methodology for each clock between different days.

More generally, we note that historical frequency ratio measurements shown in Fig. 3(a) (and extended in Fig. 5 of End Matter) exhibit disagreement at the few $\times 10^{-16}$ level, albeit with significance at the 2σ – 4σ level rather than 14σ . Additional repeated comparisons—both between JILA and NIST and at other laboratories—are necessary to properly resolve these discrepancies and set the stage for redefinition of the second. With a larger dataset, sophisticated meta-analysis methods [64–66] can be employed to build consensus among measurements overscattered relative to their individual uncertainties.

Clock network analytics—Simultaneous comparison of three different clocks gives access to analytical tools not available for individual frequency ratio measurements. Comparing sets of multiple ratios, we are able to identify that the main discrepancy between comparison campaigns likely originates from a change in the ^{87}Sr frequency or optical network [67]. Additionally, by restricting the dataset to only times when all three clocks are operating, we can analyze aspects of individual clocks and ratios, as shown in Fig. 4.

Figure 4(a) compares the normalized ratio results in pairs from each day of comparison. This demonstrates the degree of correlation in statistical fluctuations of the ratios, which helps understand individual clock and network behavior. On each subplot, one clock is common to both axes; the

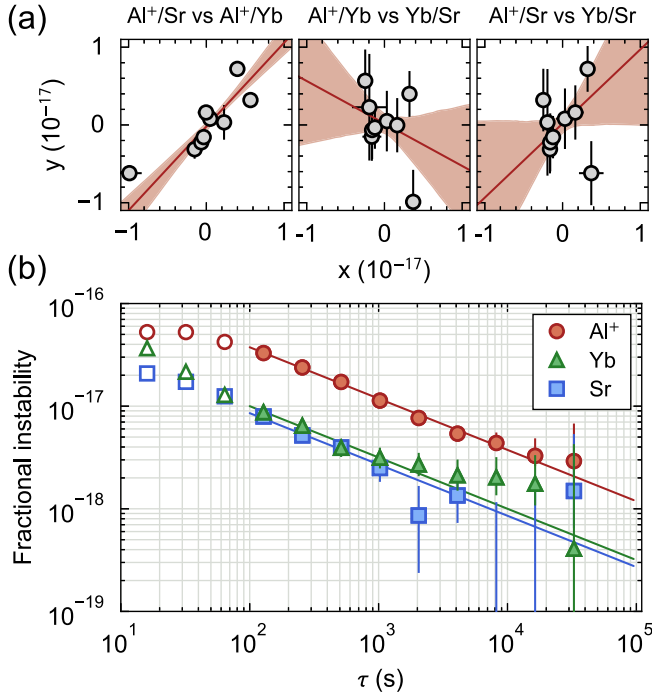


FIG. 4. Insights making use of three clocks running simultaneously. (a) Correlations between day-to-day fractional variation in pairs of frequency ratios (y vs x). Each data point is one measurement day, restricted to times when all three clocks operated simultaneously. The mean of all nine days is subtracted from each data point. The Al^+/Yb and Al^+/Sr ratios are strongly correlated, as their fluctuations are dominated by single-ion quantum projection noise. The Yb/Sr ratio shows no statistically significant correlation with either of the ratios involving $^{27}\text{Al}^+$. Red lines and shaded areas are a linear fit to the data and 1σ uncertainty, with slope given by $1.08(18)$, $-0.55(76)$, and $0.95(97)$ for the three subplots. (b) Estimating the fractional frequency stability of the three individual clocks. Points show overlapping Allan deviations for each clock, calculated using a three-cornered-hat analysis. Solid lines are weighted white noise model fits to data points with averaging time $\tau > 100$ s (filled markers).

statistical uncertainty from that clock is therefore common. To determine the statistical uncertainties from each individual clock, we perform a three-cornered-hat analysis [68]. As expected, the Al^+/Yb and Al^+/Sr ratios are strongly correlated, since statistical fluctuations in both ratios are dominated by common-mode statistical fluctuations due to QPN in the single-ion $^{27}\text{Al}^+$ clock. Conversely, the Al^+/Yb and Yb/Sr ratios show no statistically significant correlation. Using three-cornered-hat analysis on the nine-day concatenated dataset, we derive stability estimates for each clock, plotted in Fig. 4(b). We note that the overlapping Allan deviation is not well defined for gapped data in the presence of nonwhite noise, complicating this analysis for long averaging times [69]. While common-mode noise from the Si cavity means the three clocks are

not statistically independent at short averaging times, at intermediate times from 100 to 5000 s the analysis gives a good approximation of the individual clock stability. Using a white noise model, we obtain fractional instabilities of $3.75(6) \times 10^{-16}/\sqrt{\tau/s}$ for $^{27}\text{Al}^+$, $1.00(6) \times 10^{-16}/\sqrt{\tau/s}$ for ^{171}Yb , and $8.5(6) \times 10^{-17}/\sqrt{\tau/s}$ for ^{87}Sr .

The correlation of the Al^+/Yb and Al^+/Sr single-day ratios confirms that ratio statistical variation is dominated by single-ion QPN. However, these analytics are most useful when all three clocks have similar stability. Despite the significant improvement in $^{27}\text{Al}^+$ stability, the single-ion QPN is still too large to identify a source of between-day variability leading to the larger-than-expected scatter in Yb/Sr ratio values. However, the $^{27}\text{Al}^+$ stability is sufficient to characterize the short-term stability of the lattice clocks—including, to our knowledge, the first demonstration of sub- $10^{-16}/\sqrt{\tau/s}$ instability (^{87}Sr) in an atomic clock with a three-corner-hat measurement. Future operation of a multi-ion $^{27}\text{Al}^+$ clock, extending the $^{27}\text{Al}^+$ probe time with differential spectroscopy [70], or extending the comparison averaging time will help identify the source of between-day variability.

We additionally calculate Pearson correlation coefficients between the ratio means using their posterior distribution of the Bayesian model [24,56]. These quantify the correlations between the final ratio values [Eq. (1)]. These include the systematic uncertainty of the shared clock, as well as correlations in statistical fluctuations. For pairs of ratios that share $^{27}\text{Al}^+$, ^{171}Yb , and ^{87}Sr , these are 0.39, -0.74 , and 0.27, respectively.

Conclusion—We measure the frequency ratios between $^{27}\text{Al}^+$, ^{171}Yb , and ^{87}Sr optical atomic clocks with fractional uncertainties $\leq 3.2 \times 10^{-18}$. All three clocks reevaluate systematic uncertainties and use a new distribution system for ultrastable light between institutes. In principle, this accuracy allows for chronometric leveling at the 3 cm level. Additionally, simultaneous operation of more than two clocks gives access to valuable information about single-clock performance and network correlations.

Between this work and ratios we measured approximately 7 years earlier [11], we observe a difference of approximately 14σ in the ratios involving the ^{87}Sr clock, and a difference of approximately 2.4σ in the Al^+/Yb ratio. Deliberate modulation of frequency-shifting systematic effects in each clock supports the accuracy evaluations at a level below these discrepancies. This time, the loopback measurement including both combs and the fiber link between NIST and JILA confirms accurate operation of all instruments common to the Al^+/Sr and Yb/Sr ratios. Repeated measurements are necessary to resolve the ratio discrepancy and to strengthen confidence in results obtained by all laboratories. The fast averaging time and validation protocols demonstrated in this Letter will enhance future measurements and ultimately help build consensus.

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Data availability—The data that support the findings of this Letter are available from the authors upon reasonable request.

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End Matter

Figure 5 presents an expanded version of Fig. 3 in the main text. Point types distinguish measurement methodologies: open symbols represent ratios between absolute frequency measurements relative to Cs standards;

gray-shaded symbols indicate ratios via microwave link; colored symbols represent direct optical comparisons. This Letter, together with BACON21 contributes the only direct measurements of the $^{27}\text{Al}^+$ ratios published to date.

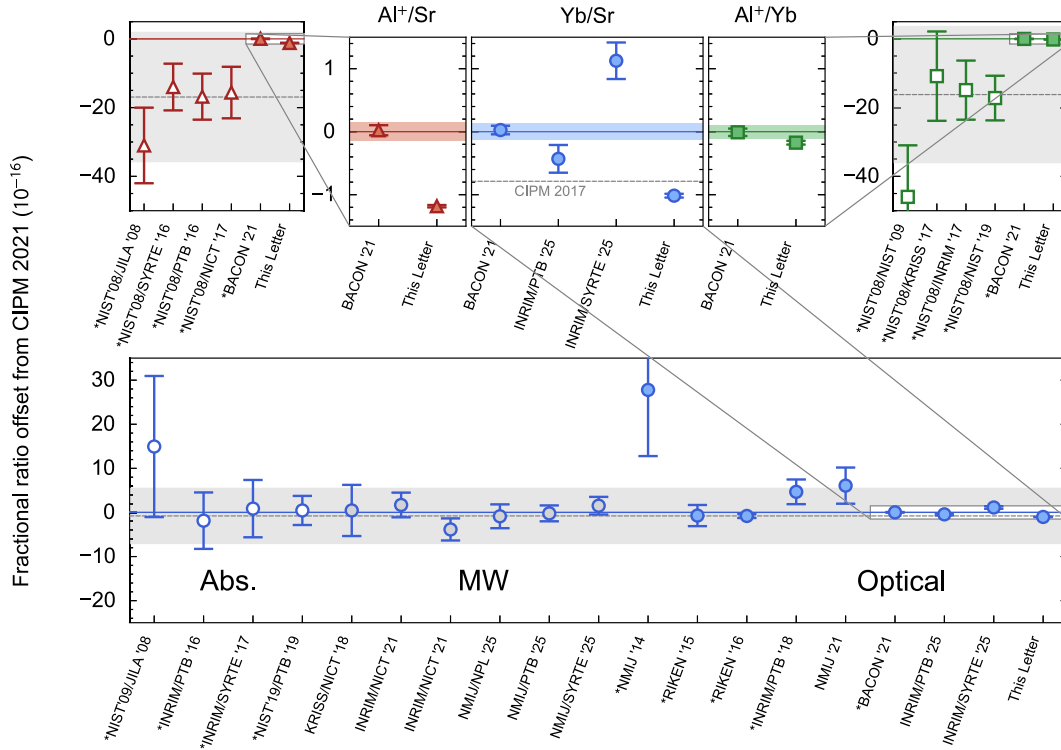


FIG. 5. Evolution of frequency ratio measurements. Asterisks indicate the values we presented in BACON21 [7,11,57,71–81]. The error bars represent 1σ uncertainty. Direct ratio measurements performed after CIPM 2017 are shown, organized by measurement method and year [58,82–84]. Colored lines and shaded regions represent the mean values and uncertainties from CIPM 2021 [56], while gray lines and shaded regions indicate the mean ratio and uncertainty from CIPM 2017 [85]. Marker styles denote different measurement methods: filled circles for direct optical measurements, open circles for absolute frequency measurements, and gray-filled circles for microwave measurements.